

BSE Group Assignments: Task Sheet

PCA Formative 2, PCA Pair Team 15

Team Members

Name	Role / Responsibility	Email
Teta Butera Nelly	Data handling, PCA implementation, plots, benchmarking, repo setup	t.nelly@alustudent.com
Mutoni Keira	Written interpretations, Colab verification, final review of the submission	m.keira@alustudent.com

Task Allocation and Tracker

Task Allocated	Assigned Member	Deadline	Completion Status	Reviewed by Team?	Comments
Choose an African dataset with missing values and text columns	Teta Butera Nelly	2026-09-30	Completed	No	OWID energy data, 54 African countries, 2000-2022.
Data handling: find missing values, impute, encode text columns, fix skew	Teta Butera Nelly	2026-09-30	Completed	No	
PCA from scratch: standardize, covariance, eigendecomposition, sort, project (Steps 1-7)	Teta Butera Nelly	2026-09-30	Completed	No	
Explained variance and choosing the number of components (Task 2 code)	Teta Butera Nelly	2026-09-30	Completed	No	
Before/after PCA plots, loadings and information-loss chart (Step 8)	Teta Butera Nelly	2026-09-30	Completed	No	
Performance benchmarking on large datasets (Task 3)	Teta Butera Nelly	2026-09-30	Completed	No	
Write the covariance paragraph and the 3 interpretation answers (max 5 lines each)	Mutoni Keira	2026-09-30	Not started	No	Placeholders are marked in the notebook.
Run the notebook in Google Colab, verify every output, and	Mutoni Keira	2026-09-30	Not started	No	

post a short verification note on GitHub (an Issue or a comment)					
GitHub: accept the collaborator invite, work on a branch and open a pull request for Nelly to review	Mutoni Keira	2026-09-30	Not started	No	
GitHub: add a "How to run in Colab" section and a results summary to the README (own words)	Mutoni Keira	2026-09-30	Not started	No	
GitHub: write data/DATA_DICTIONARY.md describing the 15 features and the added sub_region column	Mutoni Keira	2026-09-30	Not started	No	
Rebuild the combined PDF with the final answers, commit it to the repo, final check, submit	Teta Butera Nelly and Mutoni Keira	2026-09-30	In progress	No	Repo created; combined PDF to be rebuilt after the written answers are added.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps